End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B.Tech. (EC) V semester
Name of the Paper: Probability Theory and Stochastic Processes
Paper Code: TEC 503

Time: 3 hour | Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. Discuss addition and multiplication laws of probability. Differentiate between mutually exclusive and independent events with suitable examples. (CO1)

b. Two, six-sided dice are rolled. Compute the probability of occurrence of following events:
(i) First die shows number 1 and second die shows number 6.
(ii) The number 6 is shown by either first die or second die.
(iii) The sum of outcomes of the two dice is 5. (CO1)

c. Consider a digital communication system which uses channel X with a probability of 0.7, and channel Y with a probability of 0.3 for transmission of bits. The probability of error in the received bit for channel X and Y are 0.02 and 0.08 respectively. If a received bit is found in error, compute the probability that it came from channel Y using Bayes' theorem. (CO1)

Q2. (2X10=20 Marks)

a. Discuss Binomial Distribution. Derive the expression for mean of it. (CO2)

b. Consider a random variable X = {-2, -1, 0, 1, 2} with probabilities $P_X(-2) = P_X(2) = 0.2$, $P_X(-1) = P_X(1) = 0.3$, $P_X(0) = 0.5$. Compute the mean and variance of X. (CO2)

c. Explain the CDF of a discrete random variable. Sketch the CDF of a discrete random variable X having the following probability distribution:

$$p(0) = P\{X = 0\} = 1/4$$
$$p(1) = P\{X = 1\} = 1/2$$
$$p(2) = P\{X = 2\} = 1/4$$

(CO2)

Q3. (2X10=20 Marks)

a. Discuss about a random variable? Differentiate between discrete and continuous random variables with examples. (CO3)

b. Discuss the uniform distribution U (a, b) of a continuous random variable X. Derive the expressions for its mean and variance. (CO3)

End Term (Odd) Semester Examination December 2024

c. Consider a continuous random variable X having following PDF:

$$f(x) = \begin{cases} 3x^2 & \text{if } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Compute (i) E[X] (ii) P (0.3 < X < 0.7) (CO3)

Q4. (2X10=20 Marks)

a. Explain the physical significance of correlation coefficient of two random variables X and Y? How it can be computed? (CO4)

b. Consider the following joint probability distribution of random variables X and Y. Compute probabilities P(X=1), P(Y=2), P(Y=0| X=2), P(X=0| Y=2). Also find E[XY].

X	Y	p(x, y)
0	0	0.1
0	1	0.04
0	2	0.02
1	0	0.08
1	1	0.2
1	2	0.12
2	0	0
2	1	0.14
2	2	0.3

(CO4)

c. Discuss joint PDF of two continuous random variables X and Y. How marginal and conditional densities can be computed from it. Explain with an example. (CO4)

Q5. (2X10=20 Marks)

a. Discuss a random process? Explain ergodicity. (CO5)

b. The autocorrelation function of a random process X(t) is given as:

$$R_X(\tau) = e^{-\alpha|\tau|}.$$

Compute the PSD of X(t). (CO5)

c, Discuss the following:

(i) Stationary random process

(ii) Central limit theorem (CO5)